

GEC-ICP trained loss: calculation sheet

Executed part_source_v1 baseline · transformed target space

1 Residual → Huber

$$e_t = \widehat{\widetilde{y}}_t - \widetilde{y}_t$$

$$H_1(e) = \frac{1}{2}e^2 \quad (|e| \leq 1)$$

$$H_1(e) = |e| - \frac{1}{2} \quad (|e| > 1)$$

2 Per-target spatial loss

$$\ell_t = L_{point} + 0.02L_{gradient}^{norm} + 0.05L_{multi}$$

$$s_{b,t,q} = \max(\text{RMS}_{\Omega_p}(\Delta_q \widetilde{y}_{b,t}), 0.05)$$

$$L_{point} = \langle H_1(e) \rangle_{\Omega_p} \quad \cdot \quad L_{gradient}^{norm} = \text{mean}_{q=r,z} \langle H_1(\Delta_q e / s_{b,t,q}) \rangle$$

$$L_{multi} = \text{mean}_{s=2,4} \langle H_1(A_s e) \rangle \quad \cdot \quad s_{b,t,q} \text{ is per case / target / direction}$$

3 Target weighting

$$L_{ICP} = 1.0\ell_{n_e} + 1.0\ell_{n_i} + 1.2\ell_{T_e} + 0.5\ell_{\phi}$$

weights are not divided by 3.7

n_e, n_i : log₁₀ + Z-score · T_e : log1p + robust · ϕ : signed-log1p + robust

structure-holdout Train only · casewise plasma mean · boundary term disabled · Physics term disabled